2a Relational comparision

Ангид

1. Дараах илэрхийллийг ашиглан хоёр тооны хамгийн ихийг олж болох бол хоёр тоог гараас авч эхлээд багыг нь дараагаар нь ихийг нь хэвлэн харуулах программ бич.

Lab_02.cpp

Lab_02 > Lab_02.cpp > ...

```c
# include <stdio.h>
int main(){
    int a, b;
    printf("a тоог оруулна уу: ");
    scanf("%d", &a);
    printf("b тоог оруулна уу: ");
    scanf("%d", &b);
    int c = (a > b)*a + (a<=b)*b;
    printf("baga too: %d\n", !c);
    printf("ih too: %d", c);
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
macbook@macbooks-MacBook-Air C_language % cd Lab_02
macbook@macbooks-MacBook-Air Lab_02 % clang Lab_02.cpp -o bod2
macbook@macbooks-MacBook-Air Lab_02 % ./bod2
a тоог оруулна уу: 15
b тоог оруулна уу: 5
baga too: 0
ih too: 15%
macbook@macbooks-MacBook-Air Lab_02 %
```

2. Дараах үйлдэл хийгдэхэд “bolj bna” гэдэг үг хэвлэхийн тулд a,b,c-д ямар утга оноож өгөх вэ?

Lab_02.cpp Angid2.cpp

Lab_02 > Angid2.cpp > ...

```c
#include <stdio.h>
int main(){
    int a=6, b=9, c=7;
    if (c < b && a < c)
    {
        printf("bolj ");
    }
    if (a%3==0||c%5==3)
    {
        printf("bna\n");
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
macbook@macbooks-MacBook-Air C_language % cd Lab_02
macbook@macbooks-MacBook-Air Lab_02 % clang Angid2.cpp -o a
macbook@macbooks-MacBook-Air Lab_02 % ./a
bolj bna
macbook@macbooks-MacBook-Air Lab_02 %
```

3. 3 тоог гараас авч хамгийн их, бага, мөн дунд талыг нь олж тогтоох программ бич. Ингэхдээ if нөхцөл шалгах үйлдэл хэрэглэхгүй. Мөн 3 утга бүгд ялгаатай гэж үзнэ.

Lab_02.cpp Angid2.cpp Angid3.cpp

Lab_02 > Angid3.cpp > main()

```c
#include <stdio.h>
#include <math.h>
int main(){
    int a;
    printf("1r too oruulna uu: ");
    scanf("%d", &a);
    int b;
    printf("2r too oruulna uu: ");
    scanf("%d", &b);
    int c;
    printf("3r too oruulna uu: ");
    scanf("%d", &c);
    int d = (a>b && a>c)*a+(b>a && b>c)*b+(c>a && c>b)*c;
    printf("ih too: %d\n", d);
    int e = (a<b && a<c)*a+(b<a && b<c)*b+(c<a && c<b)*c;
    printf("baga too: %d\n ", e);
    int f = (b<a && a<c || c<a && a<b)*a+(a<b && b<c || c<b && b<a)*b+(a<c && c<b || b<c && c<a)*c;
    printf("dund: %d\n", f);
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS zsh - Lab_02

```
macbook@macbooks-MacBook-Air C_language % cd Lab_02
macbook@macbooks-MacBook-Air Lab_02 % clang Angid3.cpp -o a
macbook@macbooks-MacBook-Air Lab_02 % ./a
1r too oruulna uu: 18
2r too oruulna uu: 12
3r too oruulna uu: 10
ih too: 18
baga too: 10
 dund: 12
macbook@macbooks-MacBook-Air Lab_02 %
```

4. a,b,c-г гараас авна. Тэдгээр нь гурвалжны талуудын урт бол уг гурвалжин адил хажуут бол “адил хажуут”, зөв гурвалжин бол “зөв гурвалжин”, элдэв талт бол “элдэв талт” гэж хэвлэх программ бич.

Lab_02.cpp Angid2.cpp Angid4.cpp

Lab_02 > Angid4.cpp > main()

```c
#include <stdio.h>
int main(){
    int a;
    printf("1r too oruulna uu: ");
    scanf("%d", &a);
    int b;
    printf("2r too oruulna uu: ");
    scanf("%d", &b);
    int c;
    printf("3r too oruulna uu: ");
    scanf("%d", &c);
    int d = (a==b || b==c || a==c);
    printf("Tegsh untsugt gurvaljin: %d\n", d);
    int e = (a==b && b==c);
    printf ("Zuv gurvaljin: %d\n", e);
    int f = (a!=b && a!=c && b!=c);
    printf("Eldev talt gurvaljin: %d\n", f);
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
macbook@macbooks-MacBook-Air C_language % cd Lab_02
macbook@macbooks-MacBook-Air Lab_02 % clang Angid4.cpp -o a
macbook@macbooks-MacBook-Air Lab_02 % ./a
1r too oruulna uu: 15
2r too oruulna uu: 25
3r too oruulna uu: 15
Tegsh untsugt gurvaljin: 1
Zuv gurvaljin: 0
Eldev talt gurvaljin: 0
macbook@macbooks-MacBook-Air Lab_02 %
```

5. n он гараас өгөхөд түүнийг өндөр жил эсэхийг шалга. Доорх 2 нөхцлийг хангаж байвал өндөр жил гэж үзнэ. (1) 400-д хуваагддаг. (2) 4-д хуваагддаг бас 100-д хуваагддаггүй.

Lab_02 > Angid5.cpp > main()

```c
#include <stdio.h>
int main(){
    int n;
    printf("On oruulna uu: ");
    scanf("%d", &n);
    if(n/400==0){
        printf("Undur jil");
    }else if (n/4==0||!((n/100)==0))
    {
        printf("Undur jil");
    }else{
        printf("Undur jil bish");
    }
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
macbook@macbooks-MacBook-Air C_language % cd Lab_02
macbook@macbooks-MacBook-Air Lab_02 % clang Angid5.cpp -o a
macbook@macbooks-MacBook-Air Lab_02 % ./a
On oruulna uu: 2008
Undur jil
macbook@macbooks-MacBook-Air Lab_02 %
```

6. Өгсөн 3 тооны хамгийн их, бага тоог тус тус ол.

Angid2.cpp Angid4.cpp Angid5.cpp Angid6.cpp

Lab_02 > Angid6.cpp > main()

```c
#include <stdio.h>
int main(){
    int a, b, c;
    printf("1r too oruulna uu: ");
    scanf("%d", &a);
    printf("2r too oruulna uu: ");
    scanf("%d", &b);
    printf("3r too oruulna uu: ");
    scanf("%d", &c);
    if (a > b && a > c){
        printf("max too: %d\n", a);
    }
    else if (b > a && b > c){
        printf("max too: %d\n", b);
    }
    else{
        printf("max too: %d\n", c);
    }
    if (a < b && a < c){
        printf("min too: %d\n", a);
    }
    else if (b < a && b < c){
        printf("min too: %d\n", b);
    }
    else{
        printf("min too: %d\n", c);
    }
    if (a == b && b == c){
        printf("Buh toonuud tentsuu\n");
    }
    if (a > b && a < c || a < b && a > c){
        printf("dund too: %d\n", a);
    }
```

Angid6.cpp

Lab_02 > Angid6.cpp > main()

```c
    }
    else if (b > a && b > c){
        printf("max too: %d\n", b);
    }
    else{
        printf("max too: %d\n", c);
    }
    if (a < b && a < c){
        printf("min too: %d\n", a);
    }
    else if (b < a && b < c){
        printf("min too: %d\n", b);
    }
    else{
        printf("min too: %d\n", c);
    }
    if (a == b && b == c){
        printf("Buh toonuud tentsuu\n");
    }
    if (a > b && a < c || a < b && a > c){
        printf("dund too: %d\n", a);
    }
    else if (b > a && b < c || b < a && b > c){
        printf("dund too: %d\n", b);
    }
    else {
        printf("ddund too: %d\n", c);
    }
    return 0;

}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
macbook@macbooks-MacBook-Air Lab_02 % ./a
macbook@macbooks-MacBook-Air Lab_02 % clang Angid6.cpp -o a
macbook@macbooks-MacBook-Air Lab_02 % ./a
1r too oruulna uu: 25
2r too oruulna uu: 20
3r too oruulna uu: 115
max too: 115
min too: 20
dund too: 25
```

Гэрт

1. 4 оронтой тоо өгөхөд азын тоо бол 1 үгүй бол 0 гарга. Азын тоо гэж эхнээсээ болон хойноосоо уншигддаг тоог хэлнэ. Жишээ нь 3443, 1221, 1111. Үүнийг if нөхцөл ашиглахгүйгээр хийнэ.

Angid6.cpp gert1.cpp

Lab_02 > gert1.cpp > ...

```cpp
#include <stdio.h>
int main(){
    int n;
    printf("4 orontoi too oruulna uu: ");
    scanf("%d", &n);
    int n1 = n / 1000;
    int n2 = (n / 100) % 10;
    int n3 = (n / 10) % 10;
    int n4 = n % 10;
    printf("%s", (n1 + n2 == n3 + n4) ? "aziin too" : "aziin too bish");
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
macbook@macbooks-MacBook-Air C_language % cd Lab_02
macbook@macbooks-MacBook-Air Lab_02 % clang gert1.cpp -o a
macbook@macbooks-MacBook-Air Lab_02 % ./a
4 orontoi too oruulna uu: 1221
aziin too%
macbook@macbooks-MacBook-Air Lab_02 % clang gert1.cpp -o a
macbook@macbooks-MacBook-Air Lab_02 % ./a
4 orontoi too oruulna uu: 1234
aziin too bish%
macbook@macbooks-MacBook-Air Lab_02 %
```

2. Таны 35 онооны шалгалтад авсан оноо бутархайгаар өгөгдсөн бол авах дүнгээ 100%-д шилжүүлж, A, B, C, D,F үнэлгээгээр хэвлэн гаргана уу.

Angid6.cpp gert1.cpp gert2.cpp

Lab_02 > gert2.cpp > main()

```cpp
#include <stdio.h>
int main(){
    float a;
    printf("avsan onoogoo oruulna uu: ");
    scanf("%f",&a);
    int score = a/35*100;
    if (score>=90 && score<=100){
        printf("Үнэлгээ: A");
    }else if (score<90 && score>=80){
        printf("Үнэлгээ: B");
    }else if (score<80 && score>=70){
        printf("Үнэлгээ: C");
    }else if (score<70 && score>=60){
        printf("Үнэлгээ: D");
    }else {
        printf("Үнэлгээ: F");
    }
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
macbook@macbooks-MacBook-Air C_language % cd Lab_02
macbook@macbooks-MacBook-Air Lab_02 % clang gert2.cpp -o a
macbook@macbooks-MacBook-Air Lab_02 % ./a
avsan onoogoo oruulna uu: 33
Үнэлгээ: A%
macbook@macbooks-MacBook-Air Lab_02 % ./a
avsan onoogoo oruulna uu: 10
Үнэлгээ: F%
macbook@macbooks-MacBook-Air Lab_02 %
```

3. Өгөгдсөн тоо тэгш эсвэл сондгой болохыг дэлгэцлээд тэгш байвал 10, 4-т зэрэг хуваагдах эсэхийг, сондгүй бол 3,7-д зэрэг хуваагддаг эсэхийг шалга.

Angid6.cpp | gert1.cpp | gert2.cpp | gert3.cpp

Lab_02 > gert3.cpp > main()

```cpp
#include <stdio.h>
int main(){
    int a;
    printf("too oruulna uu: ");
    scanf("%d", &a);
    if (a%2==0){
        printf("ene too tegsh too\n");
        if (a/10 && a/4){
            printf("ene too 10,4-t huvaagdana.");
        }else{
            printf("ene too 10,4-t huvaagdahgui.");
        }
    } else{
        printf("ene too sondgoi too\n");
        if (a/3 && a/7){
            printf("ene too 3,7-t huvaagdana.");
        } else{
            printf("ene too 3,7-t huvaagdahgui.");
        }
    }
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
macbook@macbooks-MacBook-Air C_language % cd Lab_02
macbook@macbooks-MacBook-Air Lab_02 % clang gert3.cpp -o a
macbook@macbooks-MacBook-Air Lab_02 % ./a
too oruulna uu: 400
ene too tegsh too
ene too 10,4-t huvaagdana.%
macbook@macbooks-MacBook-Air Lab_02 % ./a
too oruulna uu: 21
ene too sondgoi too
ene too 3,7-t huvaagdana.%
macbook@macbooks-MacBook-Air Lab_02 %
```

4. Өгөгдсөн 10 оронтой бүхэл тоонд ямар цифр хамгийн олон орсныг ол.

Angid6.cpp | gert1.cpp | gert2.cpp | gert3.cpp | gert4.cpp

Lab_02 > gert4.cpp > main()

```c
#include <stdio.h>
int main() {
    long long a;
    int count[10] = {0};
    printf("10 orontoi buhel too oruulna uu: ");
    scanf("%lld", &a);
    while(a != 0) {
        int digit = a % 10;
        count[digit]++;
        a /= 10;
    }
    int max = count[0];
    int maxDigit = 0;
    for(int i = 1; i < 10; i++) {
        if(count[i] > max) {
            max = count[i];
            maxDigit = i;
        }
    }
    printf("Hamgiin olon orson tsifr: %d (%d udaa)\n", maxDigit, max);
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
macbook@macbooks-MacBook-Air C_language % cd Lab_02
macbook@macbooks-MacBook-Air Lab_02 % clang gert4.cpp -o a
macbook@macbooks-MacBook-Air Lab_02 % ./a
10 orontoi buhel too oruulna uu: 1122233333
Hamgiin olon orson tsifr: 3 (5 udaa)
macbook@macbooks-MacBook-Air Lab_02 % 3
```

2b Loop Ангид

1. n тоог хэрэглэгчээс уншин n удаа “programming in c” гэдэг үгийг хэвлэж хар.

Angid6.cpp | gert1.cpp | gert2.cpp | gert3.cpp | gert4.cpp | Angid2-1.cpp

Lab_02 > Angid2-1.cpp > main()

```c
#include <stdio.h>
int main(){
    int n;
    int i;
    printf("Too oruulna uu: ");
    scanf("%d", &n);
    for(i=1; i<n; i++){
        printf("programming in c ");
    }
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
macbook@macbooks-MacBook-Air C_language % cd Lab_02
macbook@macbooks-MacBook-Air Lab_02 % clang Angid2-1.cpp -o a
macbook@macbooks-MacBook-Air Lab_02 % ./a
Too oruulna uu: 10
programming in c programming in c programming in c programming in c programming in c programming in c programming i
n c programming in c programming in c %
macbook@macbooks-MacBook-Air Lab_02 %
```

2. 1-ээс n хүртэлх тэгш тоонуудын нийлбэр ол.

Lab_02 > Angid2-2.cpp > main()

```cpp
# include <stdio.h>
int main(){
    int a;
    int i;
    int sum=0;
    printf("Too oruulna uu: ");
    scanf("%d", &a);
    for (i=1; i<=a; i++)
    {if (i%2==0)
        {
        sum=sum+i;
        }else{}
    }printf("Niilber: %d\n",sum);
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
macbook@macbooks-MacBook-Air C_language % cd Lab_02
macbook@macbooks-MacBook-Air Lab_02 % clang Angid2-2.cpp -o a
macbook@macbooks-MacBook-Air Lab_02 % ./a
Too oruulna uu: 10
Niilber: 30
macbook@macbooks-MacBook-Air Lab_02 %
```

3. N тооны цифрүүдийн нийлбэрийг ол.

Lab_02 > Angid2-3.cpp > ...

```cpp
#include <stdio.h>
int main() {
    long long a;
    int sum = 0;
    printf("N too oruulna uu: ");
    scanf("%lld", &a);
    while(a!= 0) {
        sum += a % 10;
        a /= 10;
    }
    printf("Tsifruudiin niilber: %d\n", sum);
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
macbook@macbooks-MacBook-Air C_language % cd Lab_02
macbook@macbooks-MacBook-Air Lab_02 % clang Angid2-3.cpp -o a
macbook@macbooks-MacBook-Air Lab_02 % ./a
N too oruulna uu: 1234
Tsifruudiin niilber: 10
macbook@macbooks-MacBook-Air Lab_02 %
```

Гэрт

1. N тооны бүх хуваагчдыг ол.

Lab_02 > gert2-1.cpp > main()

```c
#include <stdio.h>
int main() {
    int n;
    printf("N too oruulna uu: ");
    scanf("%d", &n);
    printf("%d toonii huvaagchid:\n", n);
    for(int i = 1; i <= n; i++) {
        if(n % i == 0) {
            printf("%d ", i);
        }
    }
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
macbook@macbooks-MacBook-Air C_language % cd Lab_02
macbook@macbooks-MacBook-Air Lab_02 % clang gert2-1.cpp -o a
macbook@macbooks-MacBook-Air Lab_02 % ./a
N too oruulna uu: 10
10 toonii huvaagchid:
1 2 5 10 %
macbook@macbooks-MacBook-Air Lab_02 %
```

2. a,b тоонуудын ХИЕХ-ийг ол. ХИЕХ гэдэг нь a,b тоог хоёуланг нь хуваах чаддаг хамгийн их тоо юм.

Lab_02 > gert2-2.cpp > main()

```c
#include <stdio.h>
int main() {
    int a, b, temp;
    printf("a toog oruulna uu: ");
    scanf("%d", &a);
    printf("b toog oruulna uu: ");
    scanf("%d", &b);
    while(b != 0) {
        temp = b;
        b = a % b;
        a = temp;
    }
    printf("HIEH: %d\n", a);
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
macbook@macbooks-MacBook-Air C_language % cd Lab_02
macbook@macbooks-MacBook-Air Lab_02 % clang gert2-2.cpp -o a
macbook@macbooks-MacBook-Air Lab_02 % ./a
a toog oruulna uu: 12
b toog oruulna uu: 18
HIEH: 6
macbook@macbooks-MacBook-Air Lab_02 %
```